



Energy-efficient and SDN-enabled routing algorithm for wireless body area networks

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ABSTRACT

Wireless Body Area Networks (WBANs) are one of the special branches of Wireless Sensor Networks (WSNs) that attract attention in various disciplines such as health, engineering, biology, and computers. Also, WBANs are an important research area that can contribute to human life and health in many ways. The Software-Defined Networking (SDN) approach is a solution that can make simpler, flexible, manageable, and more efficient the heterogeneous and complex network structures, such as WBANs. In the Software-Defined WBAN (SD-WBAN) architecture, it is of great importance to extend the network lifetime due to the limited energy of the sensor nodes and the routing process is one of the most important energy consumption processes of SD-WBANs. For an effective and efficient routing approach, service quality requirement parameters, for example, throughput, energy efficiency, end-to-end delay, and packet transmission rates need to be considered. In this study, a new energy-efficient and SDN-enabled routing algorithm (ESR-W) has been developed with the use of the Fuzzy-based Dijkstra technique. So, the most appropriate route determination is performed with a central and reactive (on-demand) approach among many SD-WBAN users. SNR, battery level and hop count metrics are used for routing decisions. In order to compare ESR-W with existing protocols AODV and SDNRouting, extensive scenarios and simulations have been performed in the Riverbed Modeler simulation software. According to the results, ESR-W has been observed very successful compared to other protocols in terms of throughput, end-to-end delay, packet transmission rate, and energy consumption.

1. Introduction

Recent technological advances and new approaches such as WBAN and low-power wireless communications make new services such as remote health monitoring (e-health) and management gradually popular [1]. WBANs consist of wireless heterogeneous sensor nodes that are intelligent, small-sized, low-cost, low-power, lightweight, wearable personal networking devices. These sensor nodes are capable of detecting/observing physiological and non-physiological parameters on or within the human body and then transmitting the parameters to a predefined target via wireless communications.

The SDN paradigm is a novel network paradigm that has become increasingly popular in recent years. The SDN essentially allows the network to be managed from a logically central point and simplifies the control and management operations by providing an abstraction of data and control planes from each other [2–4]. The SDN has been originally designed for wired networks, but it has recently been adapted to different network architectures such as data center [5], cloud [6], and wireless networks [7]. Nowadays, new approaches such as SDN, IoT, and 5G are expected to increase the usage of next-generation

WBAN technologies and to contribute to the industrial and daily life networking solutions.

SD-WBAN nodes have a heterogeneous network structure and limited energy supplies. These sensor nodes, which can also be placed in the human body, cannot be charged and replaced in most cases. The IEEE 802.15.6 standard has been evolved to address service differences in short-range communication among light-weight devices surrounding the human body. It defines the physical and data link layers with one-hop and two-hop star topology. The SD-WBAN architecture using the 802.15.6 standard consists of only one coordinator node that is called HUB in the standard and several sensor nodes connected to it. Intra-WBAN communication refers to the communication between the coordinator node and the different number of sensor nodes connected to it, and Inter-WBAN communication refers to the communication between the coordinator nodes, controller and target (sink or gateway). The 802.15.6 standard proposes solutions for intra-WBAN communication only [8] but does not develop any solution for inter-WBAN communication. However, WBAN architecture is not just intra-WBAN. In this context, new approaches are needed for inter-WBAN communication.

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Most of the research on the WBAN architecture addresses medium access control (MAC) and IEEE 802.15.4 in the physical and data link layers. It is seen that the SDN approach is integrated only with WSNs and 802.15.4 protocol is mostly discussed [9–12]. However, the 802.15.4 standard is not suitable for the WBAN architecture for many reasons, including the limited data rate and high energy consumption compared to the 802.15.6 standard [13]. The 802.15.6 standard is designed to address different priorities and specifications of WBANs. Therefore, all solutions to be proposed for the WBAN should be based on the 802.15.6 standard.

Designing an effective routing protocol with the 802.15.6 standard will increase network performance [1,13,14]. However, although efforts have been made utilizing duty-cycling techniques to preserve energy consumption [8] and using energy harvesting techniques [13,15] to produce energy naturally, delays in packet transmissions and problems with energy efficiency are still encountered. In this context, ESR-W has been designed in accordance with the 802.15.6 standard in order to solve the aforementioned problems.

Many protocols have been designed in the literature for routing [16–19], which is one of the most important energy consumption processes. However, the limited energy of the SD-WBAN nodes is a critical issue which makes the use of this technology less desirable. SD-WBANs have a different topology due to their group-based mobility rather than node-based mobility. However, SD-WBAN has more strict energy restrictions in terms of transmission power than traditional WSN and Ad Hoc networks. Especially for implant nodes placed in the human body, the battery changes of these nodes may be quite uncomfortable and in some cases even require a medical operation. Therefore, the SD-WBANs should be a longer network lifetime. The routing protocols designed for wireless networks and WSNs are not suitable for the SD-WBANs because of the aforementioned problems and the unique characteristics of the SD-WBANs. Furthermore, these routing protocols have high energy consumption as a result of the routing process [20,21], and multi-hop communication is especially recommended for low energy consumption.

In multi-hop routing protocols, the next node selection process as a forwarding node (or intermediate node) is the most important step. Existing routing protocols propose various techniques for selecting the next node. However, most of these protocols try to choose the shortest path instead of the most suitable path. Thus, the resulting path can lead to higher energy consumption. In this study, energy-efficient and SDN-enabled routing protocol (ESR-W) is proposed for the SD-WBAN. The most important innovation is the consideration of SNR, battery level, and hop count metrics with the Fuzzy-based Dijkstra algorithm in ESR-W, which determines the optimal path by the controller (central node). The most suitable path is transmitted by the controller to the coordinator nodes on the path by means of flow tables.

As a result, new approaches for supporting both intra-WBAN and inter-WBAN communications with utilizing duty-cycling techniques, energy efficiency, and routing mechanism for WBAN environments are handled with the help of SDN paradigm. The authors attempted to cope with these problems in this article.

The WBAN architecture has a simpler topology than the WSN architecture. Therefore, there are many studies in the literature that say that central and reactive network approaches are more appropriate for WBANs. The controller can perform central routing and time slot allocation based on the information obtained from the network. In the routing decision stage, the most appropriate routes from each HUB node to the gateway (or access point) are selected. The SNR, hop count, and battery level metrics, which represent link quality, play a crucial role in WBAN's energy efficiency. HUB nodes that are out of gateway range are able to save energy by using multiple hops.

The main contributions of this study to the literature can be summarized as follows:

- IEEE 802.15.6 [8] for intra-WBAN and HUBsFlow [4] interface protocol for inter-WBAN for the SD-WBAN architecture are modeled with the Riverbed Modeler simulation software [22]. In this way, contrary to many studies in the literature, the WBAN architecture (both intra-WBAN and inter-WBAN) is considered as a whole and an energy efficiency solution is produced.
- A new central, on-demand and energy-efficient routing algorithm running on the controller named ESR-W is proposed for the SD-WBAN architecture. ESR-W is able to determine new-paths in real-time according to changes in the status of coordinator nodes.
- With the ESR-W, Fuzzy-based Dijkstra algorithm is used as the path cost calculation model [23].
- In addition, the ESR-W algorithm is compared with AODV [24] and SDNRouting [14] protocols for more comprehensive performance analysis.

The rest of the paper is organized as follows: Section 2 discusses the related works. Section 3 gives the details of the ESR-W protocol, including IEEE 802.15.6, data structures, flowchart, and efficient minimum-cost routing with fuzzy logic. The simulation scenarios and analyzes performed on the Riverbed Modeler simulation program are given in Section 4. Finally, the results of the study and recommendations for future studies are given in Section 5.

2. Related works

Although there are plenty of studies that pertain to WSN and SDN in the literature, few examples of studies that consider SD-WBAN architecture can be found. These studies are summarized below. All of these studies, except our previous study, HUBsFLOW [4], did not use the 802.15.6 standard with SDN. The other contribution of our study is the energy-efficient routing protocol developed by using a multi-metric Fuzzy-based Dijkstra technique.

Cicioğlu and Çalhan [4] developed a three-layer SDN-enabled WBAN architecture using 802.15.6 standard for the first time in the literature. In this study, a network interface protocol called HUBsFLOW has been developed for inter-WBAN communication and a manageable, flexible, and energy sensitive structure has been realized for SD-WBAN architecture. The controller mechanism introduced by the SDN paradigm takes over all network-related management and control operations in the proposed SD-WBAN architecture. HUBsFlow defines the standards for communication between coordinator nodes, controller, and target (sink, gateway).

In addition, Cicioğlu and Çalhan [14] proposed a novel energy-aware routing algorithm called SDNRouting for healthcare architecture. The SDNRouting algorithm takes into account the SNR and battery level metrics. In this study, the battery level is considered only as a threshold value (2J). Nodes whose battery level is below this threshold are eliminated when determining the optimal route. The difference between SDNRouting and ESR-W is that the hop count metric is also taken into consideration in addition to SNR and battery level. Also, in the ESR-W, the Fuzzy-based Dijkstra technique is used for path cost calculation model.

Luo et al. [9] proposed Software-Defined WSN architecture. With their approach called Sensor OpenFlow, they proposed various solutions for the proposed architecture. In their study as an approximation, the network packet processing and various addressing types defined for WSNs are supported. However, a real application of this study and performance evaluation in the simulation environment could not be seen.

Bera et al. [11] suggested an SDN-compliant WSN architecture called Soft-WSN. The proposed architecture includes Sensor Device Management and Topology Management components. These components also work on a controller designed with the SDN approach. The proposed approach makes it easier for users to control their devices on the network. In order to activate the device control mechanisms, the sensing task in a sensor node, detection delay, and active sleep states

are taken into account. The network topology is controlled by network management policies that can be changed at run-time to tackle with the dynamic requirements of IoT. The experimental results of Soft-WSN, which has been designed as a real application, have been shown to give very good results compared to the traditional WSN approach. Although the study is very useful for WSNs, it does not support the IEEE 802.15.6 standard, which is more suitable for WBAN architecture.

Al-Hubaishi et al. [10,25] suggested SDN-based wireless sensor and actuator network (WSAN) architecture that has a novel routing discovery mechanism. The routing decision method was able to change the current path during data transmission using the Fuzzy-based Dijkstra algorithm. So as to accomplish improved performance results for WSAN systems, an interface protocol named WSANFlow, which was responsible for all communications between the SDN controller (SDNC) and the SDN-enabled end devices, was proposed in [26]. The proposed SDNC was able to implement all management and network control procedures. Consequently, the SDN controller could optimize the instructions to be delivered to the end devices for manageable and efficiency by utilizing the WSANFlow interface protocol. Silva et al. [27] proposed an SD WBAN network architecture with three-tier. They did not give any performance analysis, only a framework approach was recommended. In our study, a novel approach is proposed by using the 802.15.6 protocol that is standardized for WBAN in the simulation environment and performance analyses are implemented in Riverbed Modeler.

As a result, there is no study that has been proposed in the literature to develop a new controller-based, on-demand, energy-efficient routing algorithm for an SD-WBAN architecture using the IEEE 802.15.6 standard for intra-WBAN and the HUBsFlow network interface protocol for inter-WBAN. Many studies have indicated that the OpenFlow protocol is not sufficient for low-power networks such as WBAN because it is not energy sensitive and has many rule-matching fields, and so new protocols have been proposed [4,9,11,28]. Some of these studies are designed for WSN networks and do not support 802.15.6. Therefore, HUBsFlow is preferred in this study as a network interface protocol designed for WBANs.

In the literature, distributed routing algorithms have been designed for WBAN architecture mostly. However, centralized routing algorithms began to attract attention thanks to the controller mechanism proposed by the SDN approach. For this reason, by comparing the developed algorithm with the most popular algorithm for WBAN architecture, it is examined whether the proposed solution is remarkable. On the other hand, it is aimed to compare centralized and distributed routing algorithms in this study. SDNRouting algorithm is a type of centralized routing algorithm, so, ESR-W has been compared with both the centralized (SDNRouting) and distributed routing algorithm (AODV). The proposed algorithm differs from many studies in the literature, as it has real-time new path calculation capabilities with the flow rules working on the controller for the WBAN architecture based on the SDN approach. In addition, there is no centralized routing algorithm developed for the Software Defined — WBAN architecture in the literature.

3. Proposed routing protocol

Energy efficiency is a very important issue for SD-WBAN architecture. Current energy-efficient routing algorithms generally use the remaining energy, transmission power, or link distance as metrics to determine the optimal path [29,30]. In our study, Fuzzy-based Dijkstra technique is used in path selection policies by using new metrics (SNR, hop count, battery level) in addition to energy efficiency for SD-WBAN architecture. The stable and appropriate path selection for the SD-WBAN coordinator nodes is performed by the controller, which has an overview of the entire network. Thanks to the centralized and reactive (on-demand) routing approach, the proposed new metrics provide more stable network connectivity and less additional route discovery (less workload).

In this study, ESR-W is designed in accordance with 802.15.6 PHY/MAC features. The 802.15.6 defines a general frame structure

Table 1

User priorities (D: Data – M: Management).

UP	Traffic	Packet type	CSMA/CA	
			CW_{min}	CW_{max}
0	Background	D	16	64
1	Best effort	D	16	32
2	Excellent effort	D	8	32
3	Video	D	8	16
4	Voice	D	4	16
5	Medical data	D/M	4	8
6	High priority medical data	D/M	2	8
7	Emergency	D	1	4

and physical and data link layer properties for communication between the coordinator node and the sensor nodes. It supports three different physical layer standards: narrowband, ultra-wideband, and human body communication band. It also supports Beacon mode with superframe as frame structure and various access phases like random access (RAP), exclusive access phase (EAP1, EAP2), the contention access phases (CAP), and management access (MAP) phases. Furthermore, this standard can also support CSMA/CA and Slotted Aloha multiple access (MA) techniques. In Table 1, the different priority values defined by this standard and the contention windows are given. In our study, 2360–2400MHz ISM band, Beacon mode with superframe frame structure, MAP-RAP access phases, and CSMA-CA MA are preferred.

RAP is defined as the access phase used between coordinator nodes and sensor nodes, while MAP is defined as the access phase used between coordinator nodes, gateway, and controller. In RAP, different priority values given in Table 1 are used with CSMA-CA MA technique recommended by 802.15.6 standard. In MAP, communication between the controller and the coordinator nodes is provided using the HUBsFlow network interface protocol, which is designed for SD-WBAN in our previous study. More detailed information on IEEE 802.15.6 [8], detailed simulation results [31] and detailed information on HUBsFlow can be found in references [4].

Fig. 1 shows the communication of ESR-W in detail. As shown in the figure, there are two different planes named the data and the control plane in the SD-WBAN architecture. In the data plane, there are coordinator nodes and sensor nodes connected to these coordinator nodes, and in the control plane, there is a controller (or controllers) with a global view of the network, which is a central node performing control and management operations. In the traditional WBAN architecture, all necessary control and management operations for wireless communication are carried out by the coordinator nodes. In the SD-WBAN architecture, the coordinator nodes manage only the sensor nodes connected to the coordinator node in the intra-WBAN architecture, and in the inter-WBAN communication, all control and management operations are transferred to the controller and the coordinator nodes are turned into the simple forwarding devices. In addition to all network control and management operations in inter-WBAN communication, the controller performs the routing decision process, which is an important source of energy consumption, for energy-efficient. As shown in Fig. 1, the controller mechanism has topology discovery, flow management, dynamic slot allocation, and routing decision units [4].

While traditional wireless routing algorithms such as Ad hoc On-Demand Distance Vector Routing (AODV) [24] find the most suitable route, all nodes calculate the optimal routes depending on the algorithm used, alternative routes are created and stored for specific periods. These processes cause more network traffic and more energy consumption. Performing these operations by the controller (central node), and assigning the most suitable path to all related nodes as reactive (on-demand), contributes greatly to the energy efficiency of the SD-WBAN architecture. In some cases, it is not possible to establish direct communication between the coordinator node and the gateway. In other words, an intermediate coordinator node is needed. This can sometimes be done with one intermediate coordinator node (one-hop)

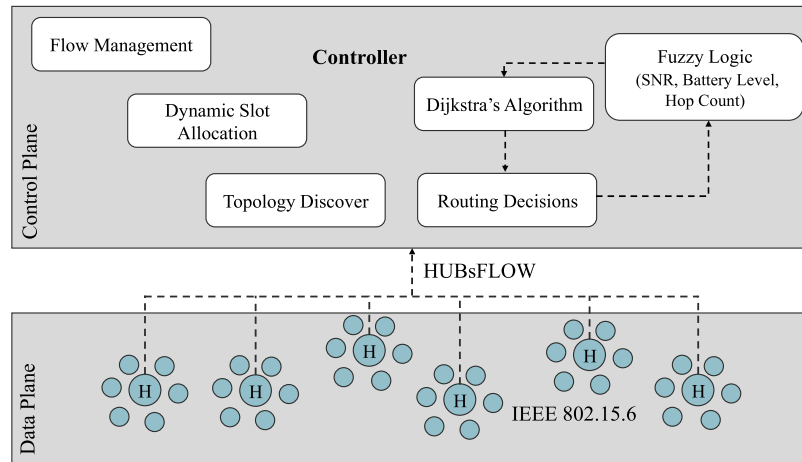


Fig. 1. ESR-W communications for SD-WBAN.

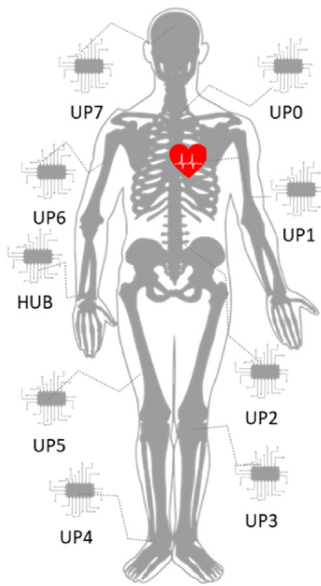


Fig. 2. IEEE 802.15.6 intra-WBAN for SD-WBAN.

and sometimes more than one intermediate coordinator node (multi-hop). In this context, it becomes a necessity to find the most suitable coordinator or coordinators to send the packet.

3.1. IEEE 802.15.6 for intra-WBAN

IEEE 802.15.6 is the standard in the SD-WBAN architecture that defines the physical and data link layers and defines the communication standards between the HUB and the connected sensor nodes. Fig. 2 shows the IEEE 802.15.6 intra-WBAN communication of SD-WBAN. In this standard, user priorities for medium access are divided into eight different access categories. The contention window (CW) values within the frame determine these priorities. These different types of traffic are classified as emergency data, high priority medical data/network control, medical data/network control, audio, video, excellent effort, best effort, and background. These different data traffics are prioritized with minimum and maximum CW values for CSMA/CA or Slotted Aloha MA techniques. In this study, CSMA/CA MA is preferred in order to realize a more realistic simulation. The minimum and maximum CW values for different user and network traffic are shown in Table 1.

Table 2

CSMA/CA-based IEEE 802.15.6 pseudo code.

```

Start: d, CWmin, CWmax, count
1: for each d do
2:   count++
3:   p ← random (0, CWmin)
4:   if medium is idle
5:     p--
6:     if (d succeeded)
7:       transmit packet
8:     else
9:       if (d failed)
10:        if (count is even)
11:          CW ← CW*2
12:        else
13:          CW is unchanged
14:        endif
15:      endif
16:    endif
17:  else
18:    pause p until medium is idle
19:  endif
20: end for

```

Table 2 provides the pseudo-code of the CSMA/CA-based IEEE 802.15.6 standard [32]. In the scenario constructed in the star topology, a saturated network traffic situation in which each node in the network sends continuous packets is handled. As given in Table 1, different priority nodes use the CW values defined for them. The sensor nodes provide access for intra-WBAN medium access by selecting a random integer evenly distributed throughout the range [1, CW] in the CW (CW_{min}, CW_{max}) range. In the CSMA/CA-based IEEE 802.15.6 standard, unlike the traditional CSMA/CA multiple access technique, the CW value doubles for an even number of errors, while keeping it unchanged for an odd number of errors. The following table defines the variable “p” for the contention value. The variable “d” is the number of attempts to access the environment.

3.2. Data structures

In the proposed approach, each coordinator node has a *neighborhood table* and a *flow table*. In the neighborhood table, each coordinator node keeps the information (SNR, hop count) of the node and its battery

level values. This information is sent to the controller if requested. SNR, hop count, and battery level values are defined between 10 dB and 30 dB, 1 hop and 15 hops and 1 J and 10 J, respectively. The controller converts the obtained information (SNR, hop count, battery level) into a cost value with the help of fuzzy logic while making the optimal path decision for the coordinator nodes. Details of the steps applied in fuzzy logic are explained in Section 3.3. Then, the shortest path algorithm Dijkstra [23] is used to find the most suitable path. In this way, none of the coordinator nodes that have inappropriate metrics are assigned as routing devices. The obtained path information as a result of the Fuzzy-based Dijkstra technique is transmitted by the controller to the requesting coordinator node and all other coordinator nodes on the path. This is done by the controller that sets and updates the flow tables in the coordinator nodes.

In the SD-WBAN architecture, each coordinator node maintains some entries in a table (flow table) which specify how to handle incoming or outgoing packets. *Flow tables* are defined in SDN-enabled coordinator nodes. Forwarding or dropping processes of the incoming packets is determined with the help of the flow tables. The coordinator nodes are only concerned with packet forwarding, which is periodically adjusted and updated by the controller. Furthermore, the new path information resulting from the routing decision is transmitted by the controller to the respective coordinator nodes via flow tables.

The controller has *Topology* and *Routing tables* to perform all of the aforementioned operations. In the *Topology table*, all neighbor table information from coordinator nodes has kept. In this table, the controller stores neighbor information and all metrics of the coordinator nodes for topology discovery. This table also serves as a knowledge base for the routing decision.

The *Routing table* contains the optimal path information calculated using the Fuzzy-based Dijkstra. As a result, the controller calculates the optimal path using the topology table, where all coordinator nodes store neighbor information and relevant metric such as battery level. Then it stores the resulting path information in the Routing table and transmits this information to all relevant coordinator nodes by means of the flow tables.

3.3. Flowchart

Fig. 3(a) shows the flowchart showing details of data transmission from coordinator nodes to the gateway. In the flowchart given in Fig. 3(b), the pathfinding process of ESR-W is detailed. The routing performed by the controller is carried out if requested by any coordinator node. So, some network traffic is eliminated which are run periodically by the routing algorithms on the coordinator nodes. As a result, energy efficiency is provided and network traffic is decreased. For example, according to the example network topology in Fig. 6, HUB1 (coordinator node) wants to send packets to the destination (gateway). First, the HUB1 looks for an appropriate flow rule in its flow table. If there is an appropriate flow rule, it forwards its packet to the next coordinator node. If there are no appropriate rules, it sends the route request packet (RREQ) to the controller for the new rule definition. After the incoming RREQ packet to the controller, the controller sends the TREQ (topology request) packet to all coordinator nodes to gather information about the network's latest status. In response to this packet, all coordinator nodes send neighbor tables (SNR, battery level, hop count) to the controller with TREP (topology response) packet. The controller updates the new network status by comparing this information with the topology table. Then, the optimal paths and, if possible, the backup paths are calculated using the Fuzzy-based Dijkstra. The most appropriate paths are placed by the controller (route reply packet — RREP) as flow rules in the flow tables of the coordinator nodes. These new paths are stored in the Routing table for reuse as needed.

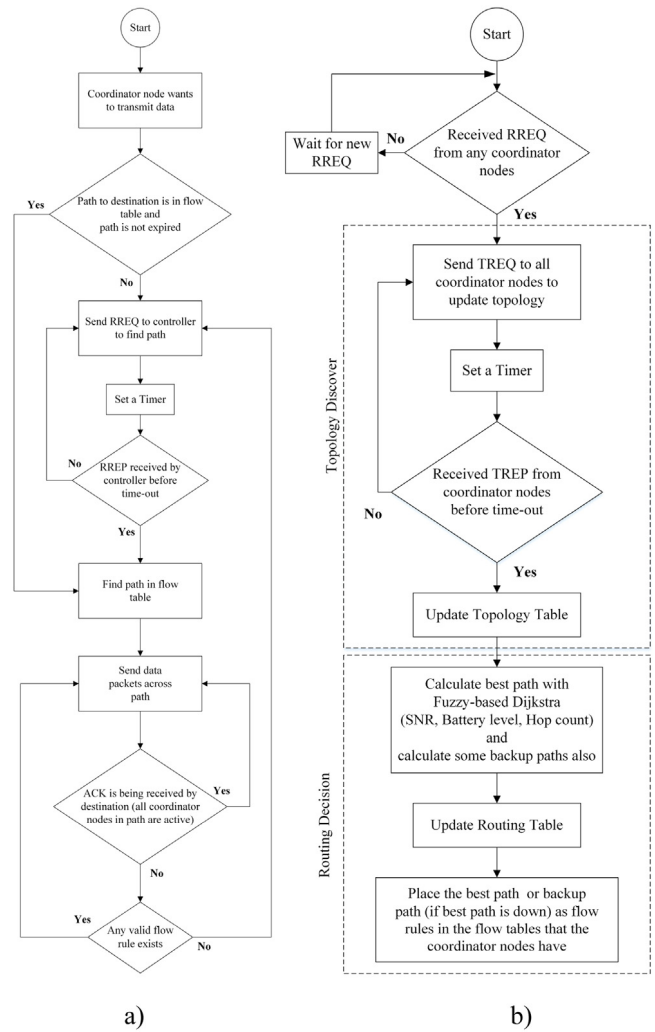


Fig. 3. (a) Flowchart for data transmission and (b) flowchart for path finding in controller.

3.4. Efficient minimum-cost routing with Fuzzy logic

Fuzzy Logic was found in 1965 by Professor Lofti Zadeh at the University of California. It is a paradigm that comprises a set of rules and procedures that we express with the boundaries and tell them what to do to be successful in solving problems within these boundaries [33]. The proposed fuzzy logic based optimal path selection process is given in Fig. 4. Three parameters named SNR, the battery level, and the hop count as crisp inputs are given to the fuzzy system. The crisp set is transformed into linguistic terms with the fuzzifier unit. The membership functions, if-then rules, and fuzzy logic operators are kept in the inference engine. The fuzzification is described as comparing task the input variables with the membership functions. Then, combining the membership values on the premise part to get the firing strength of each rule is executed. And then, qualified results are produced. The defuzzification is described as aggregating the qualified consequents to produce a crisp output and it is the last step of the fuzzy system. The knowledge base holds a number of if-then rules.

In the proposed fuzzy system, used input signal-to-noise ratio (SNR), battery level, hop count, and output cost membership functions are given in Fig. 5. Each parameter has a membership function in the developed fuzzy structure. The fuzzy input is converted to the fuzzy output using If-then fuzzy rules.

Each membership function has three functions and a total of 27 rules are stored in the rule base. Some of the rules are shown in Table 3.

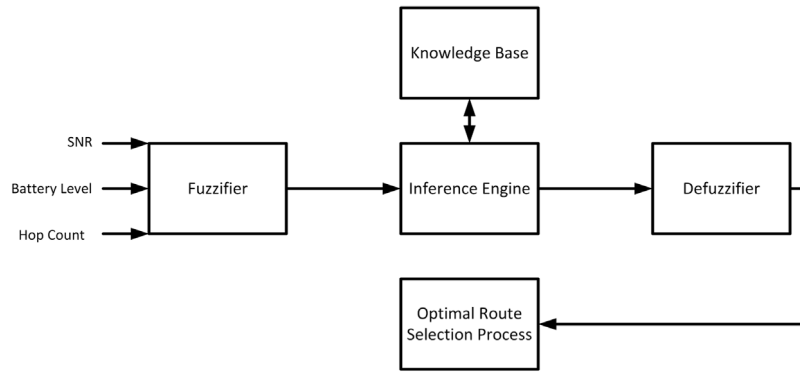


Fig. 4. Fuzzy logic based optimal route selection process.

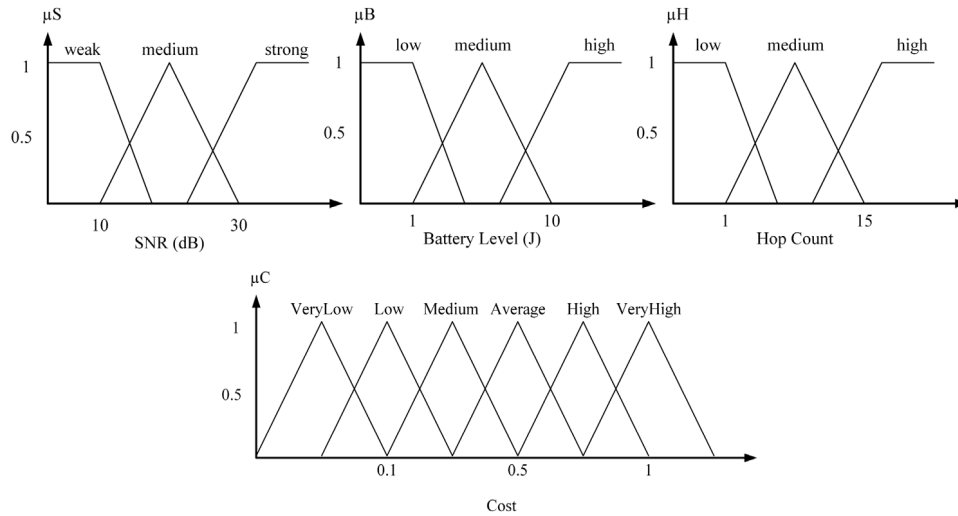


Fig. 5. Fuzzy input signal-to-noise ratio (SNR), battery level, hop count, and output cost membership functions.

There are 10 coordinator nodes and each coordinator node has the SNR, the battery level, and the hop count values in our study. The parameters are applied to the input of the proposed fuzzy decision system, and then the system produces the relevant outputs.

Fuzzy Logic is one of the most widely used techniques for multi-attribute decision-making in terms of dynamic network conditions. In this context, most of the studies on routing algorithms in literature have benefited from the Fuzzy Logic technique. Dijkstra produces very useful results in terms of the shortest path finding problem. In this context, in the ESR-W algorithm, firstly, the costs related to the paths are determined with the help of Fuzzy Logic by considering SNR, Hop Count and Battery Level metrics. Dijkstra and Bellman–Ford algorithms are the two most commonly used algorithms in the literature. In the Bellman–Ford algorithm, only the distance between neighboring nodes is taken into consideration in the distance decision process (distance vector). However, in our study, different metrics for SD-WBAN architecture have been discussed and costs have been determined with the distance. In the ESR-W algorithm, the Fuzzy Logic technique is used for connection costs and the Dijkstra algorithm provided the least costly path.

4. Simulation results

The details and results of the simulation scenarios that we perform to evaluate the performance of ESR-W are presented in this section. In this study, Riverbed Modeler software is used for a more realistic performance evaluation. It provides a simulation environment with hierarchical modeling layers and protocols for wired/wireless

Table 3
Some rules in the proposed system.

Rule no	SNR	Battery level	Hop count	Output
6	Weak	Medium	High	Verylow
13	Medium	Medium	Low	Average
14	Medium	Medium	Medium	Medium
15	Medium	Medium	High	Low
22	Strong	Medium	Low	High
25	Strong	High	Low	Veryhigh

devices [22]. ESR-W is designed with a C-Proto programming language. The AODV routing algorithm is originally designed in the Riverbed Modeler simulation program and described in detail in [34]. Information about SDNRouting is described in Section 2. Detailed information can be found in [14].

AODV is a routing protocol considered for wireless and mobile ad hoc networks and plans the routes to destinations on demand and provides both unicast and multicast routing. The AODV protocol establishes the routes between wireless nodes only if they are demanded by the source nodes, and it does not generate any extra traffic for communication along with wireless links. AODV is a significant routing algorithm developed on the distributed routing approach. The most important reason for comparing AODV routing algorithm and ESR-W algorithm is the performance comparison between distributed routing and central routing approaches for SD-WBAN architecture.

SDNRouting is a reactive and energy-sensitive routing algorithm proposed for SD-WBAN architecture. It operates according to the principle of conducting the routing decision by the controller which is

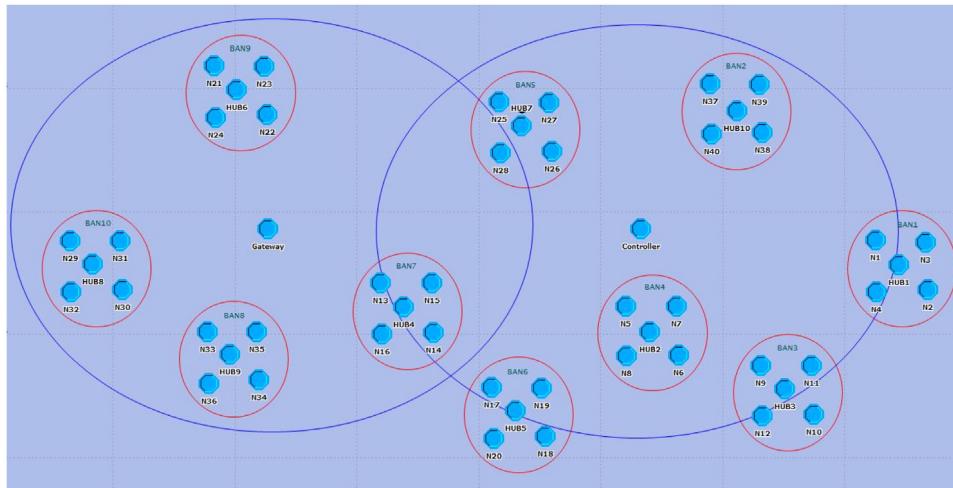


Fig. 6. Node placement in the network (52 nodes).

Table 4

Comparison of characteristics of ESR-W and AODV routing algorithm.

Performance parameters	ESR-W	AODV
Time complexity	$O(n \log n)$	$O(n)$
Multiple route possibilities	Yes	No
Route maintained in	Flow table	Route table
Routing metric	SNR, battery level, hop count	Hop count
Scalability	Good	Bad
Adaptability to topological changes	Strong	Weak
Energy consumption	Low	High

a central node and assigning the most suitable route to all relevant nodes if requested. SNR and battery level metrics are considered in the routing decision. The neighborhood table of each HUB node contains SNR data of its neighbors. In addition, the energy level states of each HUB node with the battery module are maintained. The data are sent to the controller as soon as the route request from the controller is made. The controller makes the optimal route decision based on the SNR data using the Dijkstra algorithm. The battery level metric is considered only as a threshold value and the nodes that reach this threshold value are prevented to be intermediate nodes.

SDNRouting algorithm is a central and reactive routing algorithm. However, Fuzzy Logic technique is not used for decision mechanism in SDNRouting routing algorithm. The reason for comparing the SDNRouting algorithm with the ESR-W algorithm is to observe the effect of Fuzzy Logic technique on the performance for SD-WBAN architecture in the routing decision stage.

In on-demand routing protocols route discovery process generally happens by flooding an RREQ through bidirectional links or it occurs by piggy-backing the RREP via flooding. Therefore, the time and communication complexity of the AODV algorithm is calculated as $O(n)$. However, in the ESR-W routing algorithm based on the Fuzzy-based Dijkstra technique working on the controller, the RREQ is transmitted to the controller, and route discovery operations are performed only on the controller located at a logically central point. In this way, the sensor nodes that have limited capacity are free from more calculations for the routing operations. This provides a method of less computational complexity and a better communication capability. The ESR-W routing algorithm is more efficient and scalable than the flooding-based AODV routing algorithm due to decrease in communication overhead. Comparative results of the two algorithms are given in Table 4.

The study is analyzed with three different scenarios. In the first scenario, a coordinator node (HUB1) outside the coverage of the gateway is expected to send packets with different priorities (UP7, UP4, UP2, and UP0) to the gateway. The most appropriate path to access the gateway is provided by the controller (explained in detail in Section 3). The

throughput and delay values of the proposed scenario are examined. In the second scenario, only the highest priority node (UP7) is expected to send packets to the coordinator node (HUB1), which is also outside the gateway coverage. The coordinator node also receives assistance from the controller to send these highest-priority packets to the gateway. In this scenario, the ESR-W, AODV [24] and SDNRouting [14] routing protocols are compared in terms of delay, successful transmission rate, and energy consumption. In the last scenario, it is assumed that the sensor nodes connected to the coordinator node have the same packet inter-arrival time values. In this scenario, the energy consumption values of different offered load values are examined for ESR-W, AODV, and SDNRouting routing protocols.

In all scenarios, the ID-centric data forwarding method is preferred and each coordinator node handles packets according to the IDs in the flow tables. A sample node placement in the network is shown in Fig. 6. As shown in the figure, there are 10 coordinators, 40 sensor nodes connected to each coordinator, 1 controller, and 1 gateway.

The priority values are classified as follows from the highest priority to the lowest priority; UP7, UP6, UP5, UP4, UP3, UP2, UP1, and UP0. UP7, UP4, UP2, and UP0 are selected from these priorities to avoid complexity in the figures resulting from the simulation. Each node as a result of a heterogeneous network structure of SD-WBAN has different data rates and different tasks with the 802.15.6. In this context, different inter-arrival time values have been used to make our scenarios more realistic. These values are given in Table 5 (UP7 has 4 pps, UP4 2 pps, UP2 1 pps, and UP0 0.5 pps data rates).

In the IEEE 802.15.6 standard, 8 different priorities are defined. With reference to these priorities, in our scenarios, each HUB node coordinates sensor nodes with 8 different priorities connected to it. Each priority is classified on the basis of quality of service requirements recommended by the ISO/IEEE 11073 personal health data standard for medical devices.

WBAN architecture has a heterogeneous network structure. In this context, different sensor priorities, data rates and different tasks are assigned to each sensor node in order to obtain more realistic simulation results. For example, the UP7 sensor node is assumed to have 72000 bps according to ECG medical data. Other sensor nodes are categorized into different medical data such as SPO2, accelerometer and glucose. The packets are produced with the exponential distribution function for a more realistic simulation environment. The medical application classes of WBANs are prepared based on ISO/IEEE 11073 and the details are given in [4,35].

MicaZ energy consumption values are generally used in the literature for WBANs, so MicaZ energy parameters are utilized with the default values given in datasheet [36]. All simulation parameters are given with details in Table 5.

Table 5
Simulation parameters.

Parameter	Value	
Simulation time	600 s	
Region	300 m × 300 m	
Frequency	2400 to 2483.5 GHz	
Number of nodes	80 nodes 10 HUBs	
MAC protocol	IEEE 802.15.6 for intra-WBAN HUBsFlow for inter-WBAN	
Bandwidth	1 MHz	
Data rate	971.4 kbps	
Packet size	100 Byte	
Packet inter-arrival time (<i>Generated using the exponential distribution function</i>)	UP7 = 0.25 s	UP2 = 1 s
	UP4 = 0.5 s	UP0 = 2 s
Initial node energy	10 J	
<i>MicaZ parameters [36,37]</i>		
Battery	2 AA Battery (3 V)	
<i>Current draw</i>		
TX, 0 dBm	17.4 mA	
Receive mode	27.7 mA	
Idle mode, voltage regular on	35 μA	
Sleep mode, voltage regulator off	16 μA	

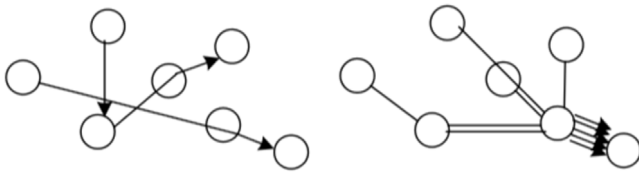


Fig. 7. The traffic patterns in ESR-W (Left: Point-to-Point Right: Convergence).

In Fig. 7, traffic patterns used in ESR-W are given. Convergence traffic pattern method is used for Intra-WBAN which allows all nodes to forward the packet to the coordinator node to which they are connected. For Inter-WBAN communication, Point-to-Point traffic pattern method is used according to the path determined by the controller.

4.1. Throughput and end-to-end delay in data transmission

One of the most important objectives in the design of ESR-W is to ensure that packets reached the gateway by selecting the most suitable paths according to the priority levels given in Table 1. At the same time, it is an important expectation that ESR-W meets service quality requirements. After receiving the path request, the controller calculates the most suitable new path. This approach is called reactive (on-demand) in routing algorithms. When the status of the coordinator nodes does not change significantly, the shortest path is also chosen by AODV and SDNRouting protocols. However, ESR-W can find new paths in any change in the status of the coordinator nodes (such as link quality, battery level, and hop count). The ESR-W selects the most suitable path optimized for the shortest path, taking into account the battery level, SNR and hop count metrics and their corresponding cost. As a result, ESR-W is an energy-sensitive approach that can take into account dynamic situations in the network and meet service quality requirements according to other protocols.

Fig. 8(a) gives the throughput of different priority packets. As shown in the figure, it is seen that the coordinator node which is out of the coverage area successfully transmits all different priority packets to the gateway. According to the obtained simulation results, different priority nodes have throughput values of 460.0459 bits/s for UP0, 861.0483 bits/s for UP2, 1698.955 bits/s for UP4 and 3565.702 bits/s for UP7. In Fig. 8(b), the end-to-end delay values of the different priority packets are given. It is observed that these different priority packets reached the target at different delay values. This is expected and desired for

the SD-WBAN architecture. Because high priority data (vital data such as heart rate) should reach the target with minimum delay. The main reason for this difference is the different contention windows defined for the different priorities in 802.15.6 (given in Table 1). As shown in the figure, it is seen that the coordinator node which is out of the coverage area transmits all packets of different priority values to the gateway with a delay of less than about 0.01 s.

In Fig. 9, the end-to-end delay and successful transmission rate results obtained in all three protocols when the packets of the UP7 node reach the gateway are given comparatively. The packets of the UP7 node are in the urgent and important medical data class. In this context, it is an important expectation for SD-WBAN that this packet will reach the target with minimum delay and packet loss rate. If a coordinator node's energy is depleted or falls below a certain threshold, this node should not be used as an intermediate node to prevent failing the network. In Fig. 9(a), it is seen that the UP7 priority packet has about 0.007 s result. Also, SDNRouting and AODV have 0.0089 s and is 0.011 s end-to-end delay, respectively. When the results are examined, it is seen that the ESR-W has the lowest delay and the AODV has the highest delay. According to the successful transmission rates of the UP7 priority packet given in Fig. 9(b), ESR-W, SDNRouting, and AODV have approximately 97%, 94%, and 90%, respectively. In this context, ESR-W has shown good results in terms of both delay and successful transmission rates compared to the other two protocols.

The design of the AODV according to the distributed routing approach and the fact that each node has an active role in the route decision stage shows a negative result in terms of delay and successful transmission rate. However, the ESR-W, designed with a central routing approach, determines the optimal path by the controller. In SDNRouting, the Fuzzy-based Dijkstra algorithm is not used and the battery level metric is considered as a threshold value, so it gives lower performance than ESR-W. Also in ESR-W, delays due to overload traffic are prevented. Another advantage of the central routing approach is that it has global knowledge about the network and therefore the most accurate decisions can be made.

4.2. Energy consumed in data transmission

ESR-W reduces the workload of coordinator nodes and ensures energy efficiency by selecting paths with higher energy levels for optimal path decisions. Fig. 10(a) gives the energy consumption results obtained for different hop count values. As can be seen from the figure, as the hop count increases, ESR-W consumes less energy than AODV and SDNRouting. In some cases, it is known that the increase in the

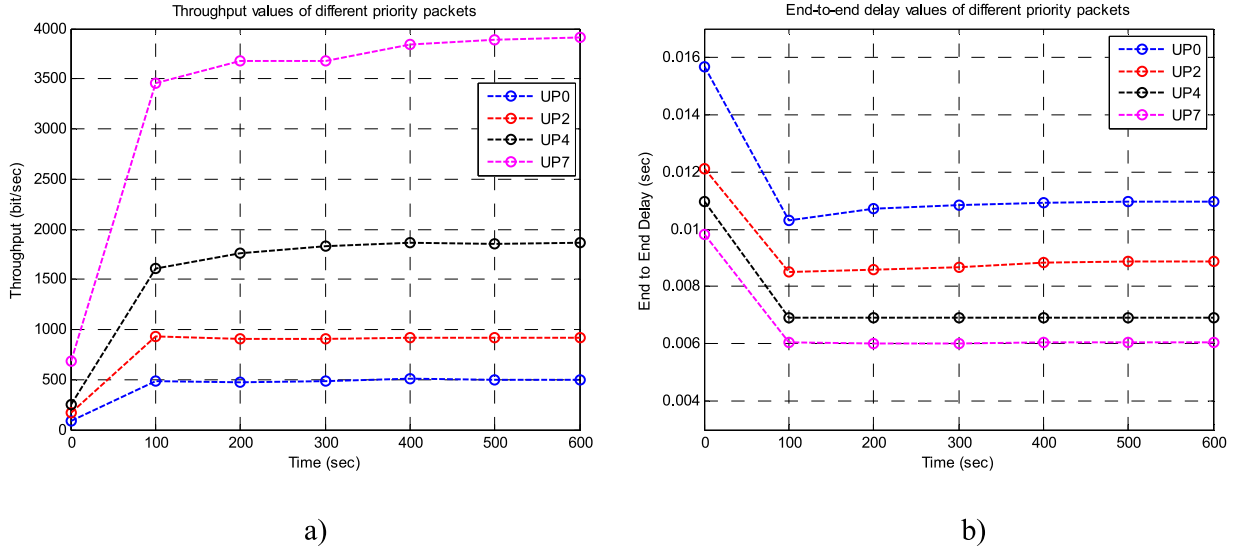


Fig. 8. (a) Throughput (b) end-to-end delay values of different priority packets.

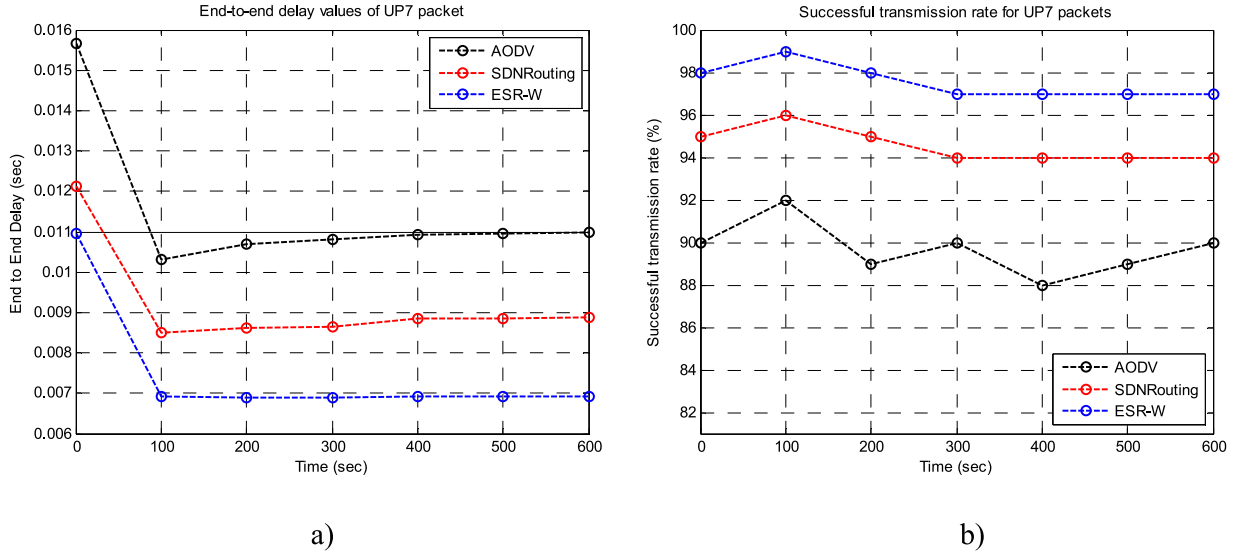


Fig. 9. (a) End-to-end delay values (b) successful transmission rate of UP7 priority packets.

hop count has a negative effect on energy consumption [38,39]. In this context, the results obtained are consistent with the literature.

Since the SNR, hop count, and battery level metrics are not considered in AODV, the optimal path between source and destination does not change each time. In SDNRouting, the battery level is evaluated only on a threshold value, and the hop count between source and target is ignored. Therefore, these two approaches seem to have more energy consumption. Furthermore, with ESR-W, the optimal path between source and destination can be changed in real-time with the help of Fuzzy-based Dijkstra technique.

Fig. 10(b) shows the energy consumption values obtained at different offered load values. For this scenario, the same packet inter-arrival time values are assigned to the priority nodes UP7, UP4, UP2, and UP0. The aim is to determine the behavior of the proposed protocol as it increases the network load. The inter-arrival time values for the nodes are selected as {0.5, 1, 2, 3, 4 and 5 seconds} respectively. The results obtained show that the energy consumption values increase as the offered load value increases as in the literature [12,21]. In a homogeneous SD-WBAN, packet size appears to have a negative impact on energy consumption. However, it has been observed that ESR-W is more successful than AODV and SDNRouting. It is seen that the

controller performs the optimal route finding process considering the current status of the SD-WBAN coordinator nodes. In this context, the proposed ESR-W is considered to be a suitable solution for SD-WBAN architecture.

The interesting and useful combination between the energy consumption and the end-to-end delay of the ESR-W algorithm in data transmission is given in Fig. 11 as delay and energy consumption trade-off. For the SD-WBAN architecture, the end-to-end delay in data transmission is returned to normal by the controller after a period of time assignment, while energy consumption is constantly increasing due to intermediate nodes and continuous data transmission. For example, end-to-end delay is returned to normal in 100. second, while energy consumption continues to increase. It is observed that the simulation period does not have a negative effect on the delay after a while and minimum consumption values continues in energy consumption. This result best demonstrates the effect of the central network approach on energy consumption with end-to-end delay.

In Fig. 12, a more intense scenario is designed for SD-WBAN architecture. In this scenario, all 8 different priority nodes (UP7, UP6, UP5, UP4, UP3, UP2, and UP1) are all addressed for each BAN. BAN8, which is outside the gateway coverage area, tries to send packets

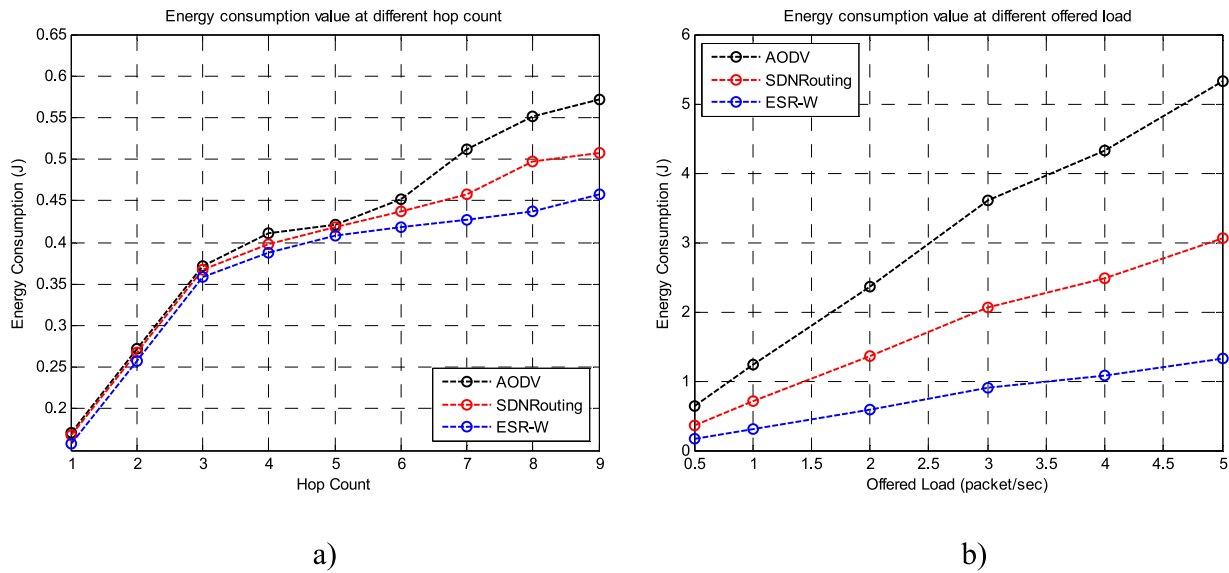


Fig. 10. Energy consumption values at (a) different hop count (b) different offered load.

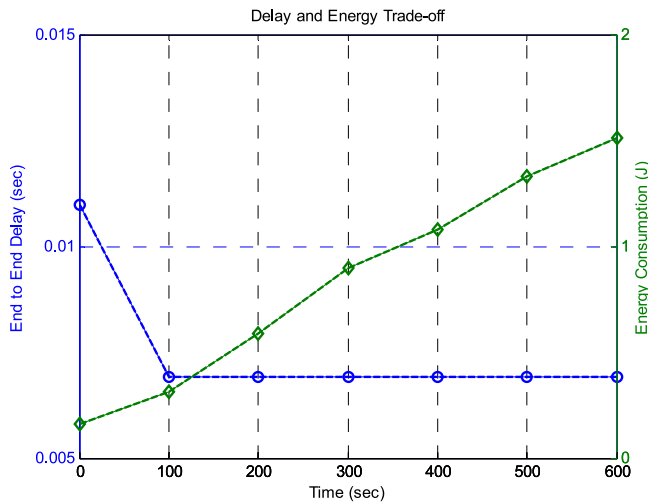


Fig. 11. Delay and energy trade-off.

received from eight different priority nodes connected to it with the ESR-W routing algorithm. There are 16 coordinators, 128 sensor nodes, a controller and a gateway (totally 146 nodes). Two different node locations are examined in this study. The node layout in Fig. 6 is referred to as scenario 1, and the node layout in Fig. 12 is referred to as scenario 2.

In our scenario with 146 nodes (16 HUBs and each HUB serves to 8 sensor nodes, so $16 \times 8 = 128$ sensor nodes are found, then add the number of sensor nodes and the number of HUBs; $128 + 16 = 144$, and then add 1 gateway and 1 controller, so $144 + 2 = 146$ nodes are totally obtained), the ESR-W routing algorithm is realized only by finding the most suitable route on 16 HUB nodes (inter-WBAN communication). In this context, maximum 16 hops can occur between the source and the destination. However, the number of hops is determined by the controller with the Fuzzy-based Dijkstra algorithm according to the defined application requirements and dynamic network conditions for different priorities (UP0-UP7). Considering the energy level of the HUB nodes, the SNR values, and the number of hops, it can make the most appropriate decisions based on different network conditions. This decision can sometimes be routes with higher energy levels and more hops. In addition, one of the most important innovations of the ESR-W

routing algorithm is the ability to set new routes in real time thanks to the SDN approach.

In Fig. 13, the energy consumption results of different offered load values for two different node locations in the ESR-W algorithm are given. As can be seen from the figure, energy consumption values increased in the second scenario. This is caused by the addition of different priority sensor nodes and coordinator nodes to the environment. In the first scenario, only 4 different priorities are addressed, while in the second scenario, 8 different priorities are used. This resulted in more offered load and energy consumption for the coordinator nodes. However, considering a dense node environment, the increase in energy consumption values is found to be normal. This result shows that new metrics used in ESR-W algorithm and Fuzzy Logic based Dijkstra algorithm provide energy efficiency in SD-WBAN architecture.

5. Conclusions

In this paper, a new energy-efficient routing protocol ESR-W is developed using the Fuzzy-based Dijkstra technique for the SD-WBAN architecture. Simulation results of ESR-W are found to be more efficient compared to popular on-demand routing algorithm AODV, and SDNRouting which is also developed for SD-WBAN architecture. In the routing decision process, using SNR, hop count, battery level metrics with Fuzzy-based Dijkstra technique gives better performance in terms of energy consumption. In AODV, it is found that some coordinator nodes are overused and some are almost never used, while ESR-W is found to solve this problem completely and partially in SDNRouting. Thanks to the SDN approach, the ability to change paths in real-time has also contributed significantly to network performance. It has been observed that ESR-W yields successful results in throughput, end-to-end delay, successful transmission rate, and energy consumption values. For future studies, it is considered that more studies will be carried out on multiple controllers and interface protocols for communication between the controllers and rule placement policies in order to use flow tables more efficiently. Since SD-WBAN architecture handles the patient's personal health data, security and privacy are important issues. In this architecture, large amounts of data are generated when data from each user are collected, thus leading to the formation of more complex systems and in turn vulnerable to cybercrime structures. In this context, in future studies, security issue will be considered for SD-WBAN architecture.

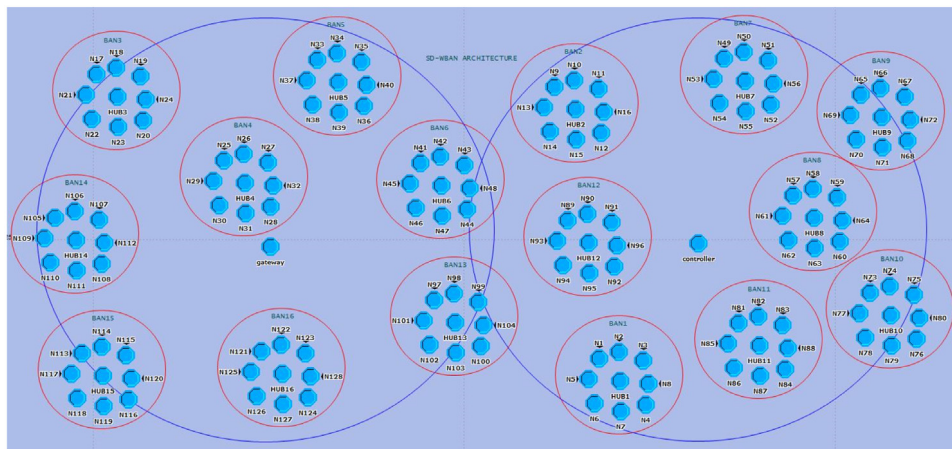


Fig. 12. Node placement in the network (146 nodes).

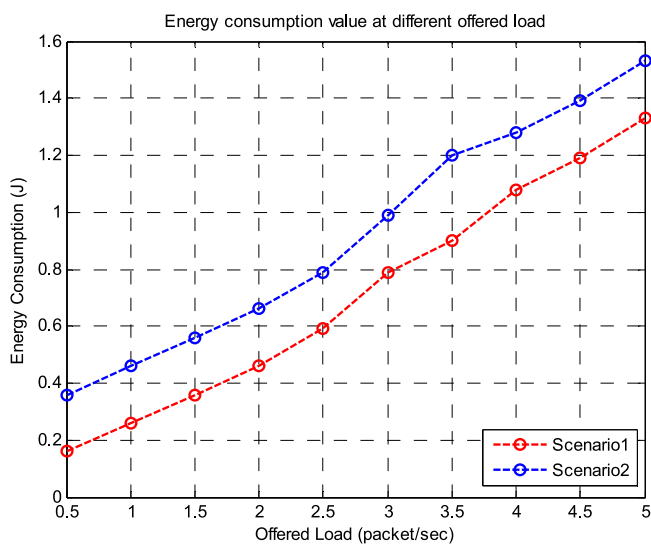


Fig. 13. Energy consumption values at different offered load for SD-WBAN.

CRedit authorship contribution statement

Murtaza Cicioğlu: Software, Investigation, Writing - original draft, Writing - review & editing. **Ali Çalhan:** Software, Investigation, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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